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Review

A review on mechanistic understanding of microplastic pollution on the performance of anaerobic digestion☆

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ABSTRACT

Anaerobic digestion (AD) has emerged as a promising technology for diverting the organic waste from the landfills along with the production of clean energy. AD is a microbial-driven biochemical process wherein the plethora of microbial communities participate in converting the putrescible organic matter into biogas. Nevertheless, the AD process is susceptible to the external environmental factors such as presence of physical (microplastics) and chemical (antibiotics, pesticides) pollutants. The microplastics (MPs) pollution has received recent attention due to the increasing plastic pollution in terrestrial ecosystems. This review was aimed for holistic assessment of impact of MPs pollution on AD process to develop efficient treatment technology. First, the possible pathways of MPs entry into the AD systems were critically evaluated. Further, the recent literature on the experimental studies pertaining to the impact of different types of MPs at different concentrations on the AD process was reviewed. In addition, several mechanisms such as direct exposure of MPs on the microbial cells, indirect impact of MPs through the leaching of toxic chemicals and reactive oxygen species (ROS) formation on AD process were elucidated. Besides, the risk possessed by the increase of antibiotic resistance genes (ARGs) after the AD process due to the MPs stress on microbial communities were discussed. Overall, this review deciphered the severity of MPs pollution on AD process at different levels.

1. Introduction

Plastics are essential materials in everyday life and their dependency is increasing because of their durability and low lifetime cost (Cucina et al., 2022; Gadaleta et al., 2022). Currently, global plastic production of ~460 million metric tons is estimated out of which ~275 million metric tons of plastic is mismanaged and released into the environment causing aquatic, terrestrial and atmospheric ecosystems (Das et al., 2021; OECD, 2022). The adverse effect of plastic pollution is mainly due to the release of microplastics (MPs) (size: <5 mm) and nanoplastics (NPs) (size: <1 μm) into the environmental system as MPs and NPs are found to be disreputably hazardous for living organisms as well as the environment (He et al., 2022; Liu et al., 2022). The last decade showed tremendous research effort on the investigation of occurrence, fate and transformation of MPs and NPs on the aquatic ecosystem along with their adverse impact on aquatic animals (Cazaudehore et al., 2022; Peng et al., 2022; Wu et al., 2022). Plastic pollution in the terrestrial ecosystems such as soil, agricultural activities, solid waste management, etc. is receiving recent attention as the MPs and NPs pollution have demonstrated to affect the sub-micro level of ecosystems such as uptake of MPs/NPs by plant cells, MPs as a career vehicle for toxic compounds such as heavy metals, antibiotics, etc. in terrestrial ecosystem (Allouzi et al., 2021; Campanale et al., 2022; Joos and De Tender, 2022). In addition, the disruption of microbial cells by the action of MPs and NPs have raised serious concerns on the biological activities in terrestrial ecosystem (Campanale et al., 2022; Mbachu et al., 2021). Among the several biological activities, organic waste management is the recent addition to the threat by possible MPs/NPs exposure causing the efficiency of treatment technology as well as becoming a secondary route of MPs/NPs entry into the terrestrial ecosystems (He et al., 2022; Tang and Hadibarata, 2021; Zhang et al., 2021a, 2021b).

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Organic waste management in urban areas possesses several challenges due to the scarcity of landfills, adoption of climate mitigation policies, and the putrescible nature of organic waste which have compelled the local authorities to adopt an efficient treatment technology for organic waste management (Chen et al., 2023; Johnravindar et al., 2021). Among the available technologies, anaerobic digestion (AD) has proved to be an efficient method to recycle the organic waste into bioenergy thereby reducing the burden on the landfills as well as the carbon emission (Johnravindar et al., 2022; Liang et al., 2021; Luo and Wong, 2019). AD technology is mostly used in urban areas to treat organic waste such as sludge and food waste individually or together as a co-digestion process for enhanced methane production (Ampese et al., 2022; Johnravindar et al., 2021; Lee et al., 2022; Liang et al., 2021). AD is a microbial-driven sensitive process wherein the sludge or food waste undergoes a series of biochemical reactions to convert the organic matter into biogas (CH_4 and CO_2) (Luo and Wong, 2019; Repinc et al., 2022). These biochemical pathways in AD process are susceptible to the environmental factors causing the microbial activities thereby affecting the AD process performance. One such environmental factor which has gained recent attention is the exposure of MPs/NPs on the AD process (Qi et al., 2021; Xu et al., 2022). The physicochemical properties of different MPs such as polyvinyl chloride (PVC), polyethylene (PE), polypropylene (PP), polystyrene (PS), polyamide (PA), polyester (PES), etc. have proven to significantly impact the AD process (Mohammad Mirsoleimani Azizi et al., 2021; Xu et al., 2022). Since the number of AD plants is increasing rapidly all over the world, it is the need of the hour to address the MPs pollution impact on AD process to develop efficient strategies for clean energy production by understanding the mechanistic insights on the role of MPs on AD process performance (Chen et al., 2021; He et al., 2021; Mohammad Mirsoleimani Azizi et al., 2021; Pereira de Albuquerque et al., 2021; Zhang et al., 2021a, 2021b).

The presence of MPs affects the AD biochemical pathways through different mechanisms. However, there are limited studies available on the mechanistic understanding of MPs on different stages of AD process. Recently, the United States Environmental Protection Agency (USEPA) has stated the major research gap existing on the impact of MPs exposure on AD process performance and has urged researchers to produce scientifically rigorous data to develop efficient AD processes (USEPA, 2021). During the AD process, the MPs can directly or indirectly affect the diverse microbial communities thereby inhibiting the process efficiency. The leaching of toxic chemicals or the formation of reactive oxygen species (ROS) from MPs during the microbial degradation process are some of the prominent mechanisms hindering the different steps of AD such as hydrolysis, acetogenesis, acidogenesis and methanogenesis (Mohammad Mirsoleimani Azizi et al., 2021; Wei et al., 2020; Wei et al., 2019a; Wei et al., 2019b). Furthermore, recent studies have demonstrated that the MPs could significantly enhance the antibiotic resistance genes (ARGs) after the AD process causing serious threats to the human health and the environment (Luo et al., 2022; Shi et al., 2022; Wang et al., 2022a, 2022b). Hence, this review aimed to provide a comprehensive assessment of the impact of different MPs exposure on the AD process pertaining to the sources of MPs entry into the AD system, the mechanistic insights of MPs effect and their influence on ARGs during AD process.

2. How do microplastics enter into anaerobic digestion system?

2.1. Sludge

The MPs generally enter the AD systems along with the feedstock materials. For instance, the sludge from the wastewater treatment plants (WWTPs) treating domestic, municipal and industrial wastewater is reported to carry a large quantities of MPs into the AD process (Fig. 1) (He et al., 2021; Liu et al., 2022; Wu et al., 2022). The global domestic and municipal wastewater is estimated as 360 km^3/year of which ~190 km^3/year (53%) is treated in WWTPs (Gao et al., 2023). The domestic wastewater includes personal care products (PCPs) such as cleansers, shampoos, cosmetics etc. containing plastic pellets (Cydzik-Kwiatkowska et al., 2022; Sun et al., 2020, 2020b). The annual MPs released from PCPs in Europe, USA and China are reported to be 3215 tons, 282 tons and 346 tons, respectively (Sun et al., 2020, 2020b). The municipal wastewater is polluted with MPs from city dust, road markings etc. whereas the industrial wastewater is reported to carry plastic pellets, synthetic textiles, industrial abrasives, etc. (Cydzik-Kwiatkowska et al., 2022; Gao et al., 2023; He et al., 2021). Overall, the WWTPs all over the world are receiving the wastewater that are highly polluted with various types of MPs ranging from 0.2 to 17,214 particles/L (Chand et al., 2021; Gao et al., 2023). The WWTPs are reported to remove more than 90% of MPs by concentrating the MPs from liquid phase into the sludge (He et al., 2021; Mohammad Mirsoleimani Azizi et al., 2021; Philipp et al., 2022). Hence, the sludge from WWTPs are highly polluted with MPs and around 30 different

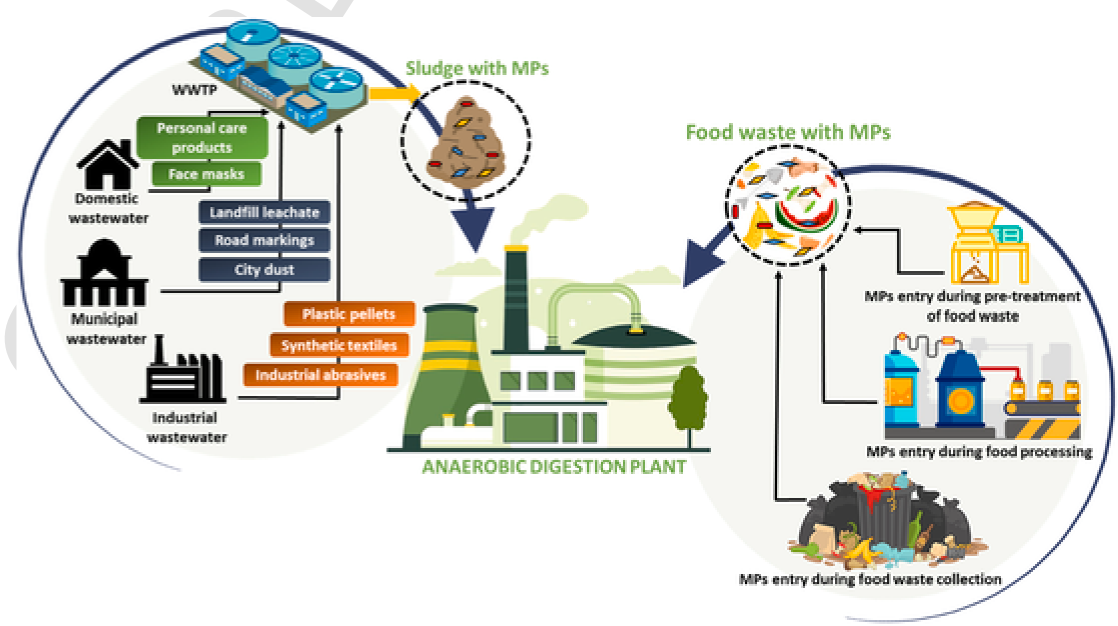

Fig. 1. Schematic representation of entry of microplastics (MPs) into the anaerobic digestion plant.

types of MPs and NPs have been reported in the domestic sludge (Wei et al., 2021; Zhang et al., 2020; Zhang et al., 2022a, 2022b, 2022c). The concentration of MPs can vary from 1.5–240 × 10^4 particles/kg dry solids (Cydzik-Kwiatkowska et al., 2022; Wu et al., 2022) in various shapes such as fibers, fragments, spheres, film, etc. (Vuori and Ollikainen, 2022). In Hong Kong, the sludge samples showed the MPs abundance of 167–936 particles/g dry sludge (Zhang et al., 2022a, 2022b, 2022c).

Traditional sludge disposal strategies included direct land application. Harley-Nyang et al. (2022) recently reported a case study showing the MPs concentration of 37.7–97.2 particles/g of dry sludge in UK. It was estimated that the annual biosolids to land applications may release 12.24 × 10^{10} to 19.32 × 10^{10} MPs which is equivalent to the same volume as 240,000 plastic bank cards. To decrease the adverse effect of direct land application, AD technology has been adopted to treat the sludge before its disposal. In UK, 73% of sludge is treated by AD process (Harley-Nyang et al., 2022), similar to the other European countries (Gianico et al., 2021). These data suggested that the large quantities of MPs are entering into the AD systems through the sludge.

2.2. Food waste

Food waste is another major organic waste treated by AD technology in urban areas for the energy recovery (Li et al., 2023; Luo et al., 2022; Manu et al., 2022). Recent studies have shown the presence of MPs in food waste streams in different countries (Porterfield et al., 2023). The MPs in food waste is reported to enter from the industrial food processing and packaging whereas the food waste collection system could also largely impact on the entry of plastics residues or MPs into the food waste (Sholokhova et al., 2021) (Fig. 1). Furthermore, generally, the food waste is collected from the source of its generation by plastic bags and transferred to the AD plants. The pretreatment of food waste is necessary to separate the plastics and food waste using mechanical ways such as shredders (Ruggero et al., 2021). There are no proper technologies available to completely ensure the separation of plastic pieces and food waste in AD plants (Porterfield et al., 2023). Hence, a large portion of MPs enter into the AD system through the pretreatment step of food waste (Golwala et al., 2021).

The level of MPs in food waste is not well reported in the scientific literature. A study by Golwala et al. (2021) showed 300,000 MP/kg of food waste (dry weight) collected from grocery stores in USA. Ruggero et al. (2021) and Schwinghammer et al. (2021) reported 1400 ± 150 MP/kg of pulped food waste (dry weight) and 40 MP/kg of homogenized food waste in Italy and Germany, respectively which were mostly comprised of PP, PE and PS. do Carmo Precci Lopes et al. (2019) and Kawecki et al. (2021) found 3–5.6% (dry weight) and 0.12% (dry weight) of MPs in household biowaste and kitchen waste in Austria and Switzerland, respectively. Apart from these, recent Covid-19 pandemic has increased the disposal of personal protective equipments (PPE) such as face mask, gloves containing PP, PU, PAN, PE, PS, etc. causing the plastic pollution in domestic wastewater as well as food waste (Benson et al., 2021; Pereira de Albuquerque et al., 2021; Shams et al., 2021).

These reports suggested that the organic waste streams such as sludge and food waste are highly contaminated with different types of MPs at different concentrations (Porterfield et al., 2023). Due to the size and persistent nature of MPs, their separation prior to the organic waste treatment is economically and technically challenging (Friege and Eger, 2021). Considering the larger interest, i.e., to divert the organic waste from landfills, the AD operators are not considering additional measures to mitigate the entry of MPs into the AD system (O'Connor et al., 2022). However, recently, the stakeholders of organic waste management such as municipal authorities, AD operators and the end product users (digestate and compost) have raised concerns about the MPs abundance in food waste as well as their residual end products after AD treatment (USEPA, 2021).

3. Impact of microplastics (MPs) on the performance of AD process

Different types of MPs play varied role on their surrounding environment due to the nature and chemical characteristics of plastics (Evode et al., 2021). Among several types, MPs such as PVC, PE, PET, PES, PS and PBS have been reported to affect the AD process (Table 1).

Table 1
Summary of studies reporting the impact of microplastics (MPs) on AD process.

MP type	MP concentration	AD Feedstock	Major findings	Reference
PVC	0.5 g/kg VS	WAS	Reduced methane production by 16.2%	Jiang et al., 2022
PVC	10-60 particles/g TS	WAS	Decrease in methane production due to leaching of BPS	Wei et al., 2019b
PE	1–4 g/L	Cosmetic industry waste	4 g/L reduced biogas production by 7% due to the adsorption of inhibitors on MPs	Akbay et al., 2022
PE	200 particles/ g TS	WAS	Reduction in methane production by 6.1–13.8% due to the reduced methanogens	Shi et al. (2022)
PE	10 mg/L	Granular sludge	Reduction in methane production by 30.71% due to the reduction in acetogens and nethanogens.	Wang et al. (2022a)
PE	10-200 particles/g TS	WAS	100 and 200 particles/g TS decreased methane production by 12.4–27.5% due to ROS synthesis and reduced enzyme activities	Wei et al., 2019a
PET	1–6 mg/g TS	Alkaline-thermal pretreated sludge	Increased methane production by 22%5 due the sludge pretreatment	Hatinoglu and Sanin (2022)
Polyester	1-200 particles/g TS	WAS	Hydrolysis and methanogenesis were inhibited	Li et al., 2020
PS	75 mg/L	Anaerobic granular sludge	Methane production was reduced by 6.7–16.2%	Wang et al., 2022b
PS	50 mg/g TS	WAS	Methane yield decreased by 15.5%	Wang et al., 2022c
PS	0.05–0.25 g/L	Glucose	0.25 g/L reduced methane production by 18–19%	Zhang et al., 2020
PS	0.05–0.2 g/L	Sewage sludge	0.2 g/L reduced methane production due to pitting of microbial cells by PS	Fu et al., 2018
PS	10–50 μg/L	Glucose	20 and 50 μg/L reduced methane production by 19–29% due to leaching of SDS and ROS generation	Wei et al., 2020
Face mask containing PE, PP, PVC	4-20 masks	Food waste	Reduction in methane production up to 18%, increase in lag phase by 7–14%	Pereira de Albuquerque et al., 2021.
PBS	0.5 g/L	Sludge	Inhibited methanogenesis by 10.2%	Tang et al., 2021

(a) PVC

PVC is one of the highly used polymer in everyday activities such as clothing, water pipes, furnitures, bottles etc. (Evode et al., 2021) and it contributes to ~18% of global plastic production (Rodrigues et al., 2019). Hence, the PVC contamination in organic waste is inevitable through various sources. The chemicals released from the PVC plastics have been categorized as highly in vitro toxic compounds, hence their pollution in the environment raises serious concerns (Zimmermann et al., 2021). Recent studies have reported the impact of PVC MPs on the AD process during sludge treatment wherein a significant inhibition on methane production was observed. Jiang et al. (2022) reported that the PVC MPs at the concentration of 0.5 g/kg VS could reduce the methane production by 16% during sludge AD process. In another study by Wei et al. (2019a), substantial reduction in methane production due to the leaching of toxic Bisphenol S (BPS) compound was observed at varied PVC concentrations (10–60 particles/g TS). The inhibition on methane production was directly proportional to the concentration of PVC MPs by directly affecting the microbial communities responsible for methane production. The high usage of PVC products at domestic and industrial level lead to the increased MPs pollution that eventually affect the AD process causing the deteriorated process performance. In addition, the profiling of toxic chemical leaching from PVC is still not clear which may bring new challenges and routes to inhibit the microbial activities in AD process (Zimmermann et al., 2021). Hence, further research pertaining to the chemical leaching and their subsequent impact on the anaerobic microbial communities is needed.

(b) PE

PE is one of the highly used polymer globally accounting for ~31% of polymer production and mainly used for packaging purposes (Rodrigues et al., 2019). Single use plastics are mostly made up of PE polymers and their mismanagement has caused severe impact on the aquatic environment. The PE contamination into the organic waste especially in food waste is easier as the food waste is mostly collected using high or low density PE plastics globally. Akbay et al. (2022) investigated the impact of two different sizes of PE (150 μm and 50 μm) MPs at the concentrations of 1 g/L, 2 g/L and 4 g/L on AD process. The results showed that 50 μm PE did not affect the methane production whereas 150 μm PE at 4 g/L reduced the biogas production by 7%. The authors suggested that the inhibition could be due to the adsorption of inhibitors on the MPs causing the reduced microbial activities. The larger surface area and the surface charge of MPs have been reported to enhance the sorption capacity of pollutants on the MPs surface thereby indirectly augmenting the inhibitory effect on microbial activities. Shi et al. (2022) reported that PE MPs with sizes 180 μm and 1 mm at the concentration of 100 mg/g TS reduced the methane production by 6.1% and 13.8%, respectively. However, the hydrolysis and acidification were enhanced in the presence of PE MPs. In another study by Wang et al. (2023), PE MPs at the concentration of 10 mg/L reduced the methane production by 30.71% due to the reduction in the microbial communities responsible for acidification and methanogenesis. In addition, the abundance of genes such as *omcB*, *mtrC*, *hupL* and *mcrA* responsible for extracellular electron transfer and methanogenesis were found to be decreased. Further, the key methanogenic enzymes such as alcohol dehydrogenase, formylmethanofuran dehydrogenase and Coenzyme-B sulfoethylthiotransferase were predicted to be decreased causing the inhibition of methane production. Wei et al. (2019b) investigated the impact of 10–200 PE particles/g TS on sludge AD process and found that 100 and 200 PE particles/g TS decreased methane production by 12.4–27.5% due to ROS synthesis and reduced enzyme activities. The mechanism of inhibition from PE MPs is still not clear, and the impact of leached compounds from PE seem to have lesser impact than the ROS synthesis which needs further investigation.

(c) PS

PS polymer is used for building, construction and packaging and ~4.4% of global production was estimated (Rodrigues et al., 2019). Wang et al. (2022a, 2022b) reported that the PS exposure at the concentration of 75 mg/L on granular sludge AD process affected the methane production by 6.7–16.2%. Similarly, Wang et al. (2022a, 2022b) observed the reduction in methane yield by 15.5% at PS MPs concentration of 50 mg/g TS during sludge AD process. Zhang et al. (2020) investigated the effect of PS MPs on the model substrate (glucose) AD process at different concentrations of PS (0.05–0.25 g/L). The lower concentrations of PS did not affect the AD process whereas the maximum concentration, i.e., 0.25 g/L reduced the methane production by 18–19%. Similarly, 0.2 g/L of PS MPs showed methane production reduction due to the pitting of microbial cells by PS exposure (Fu et al., 2018). Wei et al. (2020) examined the effect of PS MPs at the concentration level of 10–50 μg/L on AD of glucose and observed that 20 and 50 μg/L reduced methane production by 19–29% due to leaching of sodium dodecyl sulfate (SDS) and ROS generation. Further research is required to assess the impact of PS MPs on the anaerobic microbial communities to understand their mechanistic insight into the inhibition process.

Li et al. (2020) observed an inhibition of hydrolysis and methanogenesis on AD process under the exposure of 1–200 PES particles/g TS. Apart from the individual MPs studies, Pereira de Albuquerque et al. (2021) studied the degradation behavior of face mask containing PE, PP and PVC during food waste AD process. The results suggested that 4 and 20 face masks reduced the methane production up to 18% and increased the lag phase by 7–14%. In addition to the inhibition studies, few studies have reported positive impact of MPs on AD process. Hatinoglu & Sanin (2022) reported that PET MPs increased the methane production by 22% at the concentration of 1–6 mg/g TS in sizes 250–500 μm during AD of alkaline-thermal pretreated sludge. This study indicated that the negative impact of MPs could be avoided by pretreating the sludge. Overall, the majority of the reported studies have demonstrated the negative impact of different types of MPs on AD process performance, however, the specific mechanism underlying the inhibitory effect need further investigation.

4. Mechanistic insight on the effect of microplastics (MPs) on AD process

4.1. Direct mechanism

The impact of MPs on the AD process can be elucidated through different mechanisms. At the beginning of the AD process, the physical presence of MPs in the AD matrix significantly influences microbial activities (Tang et al., 2022). For instance, the MPs have been found to disrupt the microbial cells via the physical attachment to the microbial cells (Fig. 2). The MPs/NPs could break the cell walls of microbial species through the biolipid layer present on the cell wall which is referred as 'cell pitting' results in the killing of microbes as well as affecting the synergistic relationship between the microbial communities thereby reducing the methane yield during the AD process. Such a phenomenon was observed by Fu et al. (2018) wherein the PS NPs at a concentration of 0.2 g/L inhibited the fermentative bacteria *Acetobacteroides hydrogenigenes* under simulated conditions. Zhang and Chen (2020) also reported the entry of NPs into microbial cells causing the damage of cell walls.

Furthermore, Wei et al. (2019a; 2019b) reported a reduction in the abundance of hydrolytic bacteria such as *Rhodobacter* sp., *Proteiniborus* sp. and methanogenic microbial communities under the influence of PE and PVC MPs. Similarly, Jiang et al. (2022) reported decrease in the abundance of acidogenic bacterial species *Acetoanerrobium* and *Mesotoga*, proteolytic bacteria *Proteiniborus* and methanogens *Methanosaeta*

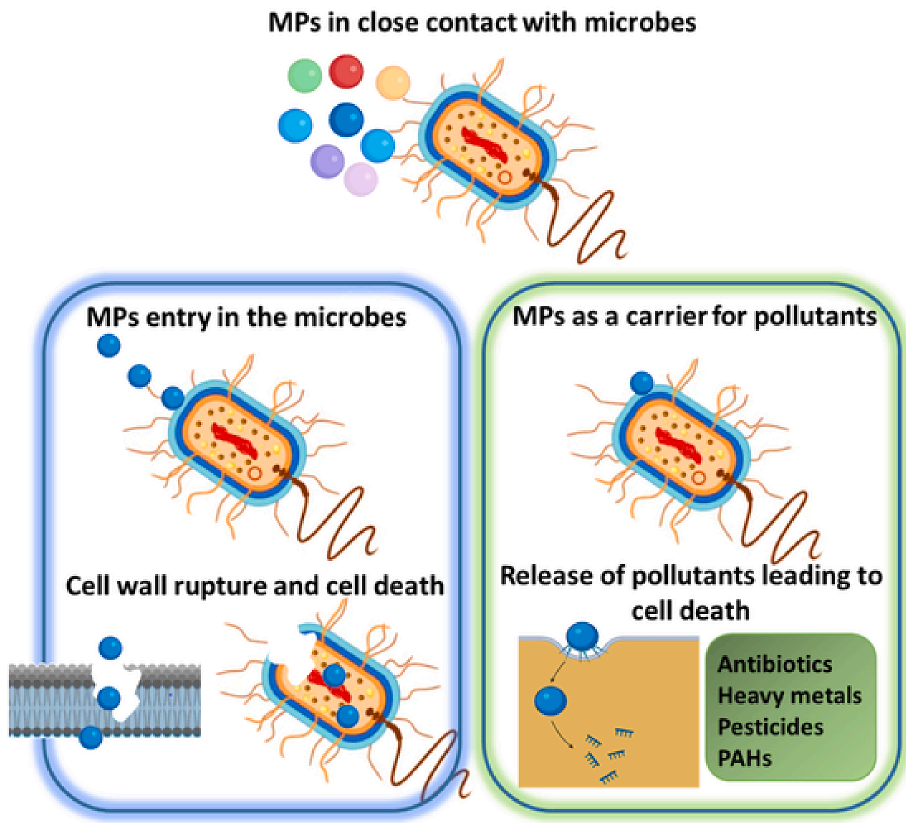

Fig. 2. Impact of microplastics (MPs) on the microbial species during AD process: Direct mechanism.

in the presence of PE and PVC MPs resulting in the decline in methane production. Besides, Li et al. (2020) have reported the inhibitory effect of PVC on *Geobacteraceae* sp. during the AD process. Apart from microbial communities, the MPs can directly influence the key enzyme activities and results in the performance deterioration. The major enzymatic activities such as protease, cellulase, kinase, coenzyme F420 etc. have found to be affected by the presence of MPs PVC and PS (Wei et al., 2019b; Feng et al., 2018b; Lin et al., 2020). Furthermore, direct toxic effect of MPs on the microorganisms have been reported in different environments indicating the potential threat of MPs on microorganisms viability (Qiu et al., 2021). Sendra et al. (2019) reported that the increase in PS concentration increased the contact area between the PS and microbes resulting in the decreased microbial density.

The size of MPs plays major role in the toxic effect on the microbial cells. The smaller MPs have reported to exert serious toxic effects on growth, oxidative stress and cell microstructure of microbes due to the increased surface area (Zhang et al., 2017). Zhao et al. (2020) reported higher negative effect on microbes by nano-level PS exposure due to physical retardation and the ability to cross the biological barrier compared to micro-level PS. Hence, size-dependent effect of MPs on the microbial community structure during AD process need further investigations.

The microbial inhibition due to the MPs exposure was mostly reported to be affected by MPs size, concentration and exposure time. For instance, PS exposure has reported to inhibit the growth rate of *E. gracilis, C. pyrenoidosa, Phaeodactylum tricornutum, Stylophora pistillata, K. mikimotoi*. Similarly, PVC exposure has affected *K. mikimoto* and *P. tricornutum*, whereas PE MPs has reported to inhibit *Scenedemussubspicatus, P. tricornutum, P. aeruginosa, B. subtilis* (Sendra et al., 2019; Qiu et al., 2021). In addition, *Phaeodactylumtricornutum* has found to be affected by the exposure of PP and PET (Song et al., 2020). In addition, the surface charge of MPs has also shown to affect the toxicity on microorganisms. The positively charged PS has reported to cause oxidative stress and membrane destruction in *cyanobacteria* (Feng et al., 2019). However, some microorganisms have the ability to overcome the MPs toxicity, hence, detailed investigations are required to develop suitable microbial inoculum for effective AD process in the presence of MPs.

These results suggested that MPs exposure has several effects on microorganisms manifesting in the destruction of cell structure and functional hindrance. Short term exposure of MPs directly affects the lipid and fatty acid composition of microbial cell walls whereas the prolonged exposure of MPs reported to damage the DNA of microbial cells. Moreover, the MPs acquaintance inhibits the synthesis of intracellular enzymes which are responsible for cell metabolism and detoxification ability (Qiu et al., 2021; Xiao et al., 2020). Hence, cell membrane lipids and intracellular enzymes can be used as markers to assess the impact of MPs exposure on different stages of AD process.

MPs are also known as the carriers of environmental pollutants in different ecosystems. The ionic properties of MPs and the large surface area assist the accumulation of environmental pollutants such as heavy metals, pesticides, antibiotics, polycyclic aromatic hydrocarbons (PAHs) etc. on the surface of MPs (Lin et al., 2022). The adsorbed pollutants on the MPs increase the exposure risk of microorganisms and the direct attachment of MPs on the microbial cell walls could create a toxic environment from the pollutants leading to microbial destruction (Fig. 2) (Qiu et al., 2021). For instance, Lin et al. (2022) observed the strong correlation between sludge MPs (PE, PET, PS, PA, PVC) and Cd adsorption with the altered microbial communities during AD process. Gao et al. (2022) reported the enhanced adsorption of antibiotic tetracycline on PP and PET MPs.

The enhanced surface reactivity of metals during biological processes such as AD increases the possibilities of metals accumulation on MPs (Qiu et al., 2021). The combination of MPs exposure and pollutant/organic matter presence causes synergistic and antagonistic toxicity to microorganisms. The PS exposure along with ibuprofen, dibutyl phthalate and tetracycline has found to cause antagonistic toxicity on *Cholrella pyrenoidosa*, *Enchytraeuscrypticus* and *Skeletonemacostatum* (Zhu et al., 2019). The PE MPs exposure with chlorpyrifos, triclosan and nonylphenol has shown antagonistic toxicity on *Isochrysis galbana*, *Skeletonema costatum* and *Chlorella pyrenoidosa*, respectively (Yang et al., 2020; Qiu et al., 2021). Similarly, exposure of PVC with triclosan induces antagonistic toxicity on *Skeletonema costatum* (Zhu et al., 2019). Apart from these, several studies have reported synergistic toxicity, for example, PS exposure with triphenyltin chloride and tetracycline has caused synergistic toxicity on *Chlorella pyrenoidosa*, *Enchytraeuscrypticus*, *Skeletonemacostatum* (Feng et al., 2020). Moreover, the metals with MPs have also shown toxic effect on the microorganisms. PS exposure with Ag, Cu, Zn and Mn has shown both antagonistic and synergistic toxicity on *E. coli*, *Tetraselmischuii*, *Chlamydomonas reinhardtii*, *Ochromonasdanica*, *Chlorella vulgaris* (Sun et al., 2020, 2020b; Qiu et al., 2021).

The pollutant adsorption capacity of MPs depends on different factors such as surface charge, hydrophobicity and surrounding environment. The increase in salinity during AD process may increase the adsorption capacity of PS with tetracycline thereby enhancing its toxicity (Feng et al., 2020). Hence, understanding the synergistic correlation between the environmental inhibitors such as MPs, antibiotics, heavy metals etc. during the AD process is necessary and demands for further research.

4.2. Indirect mechanism

(a) Leaching of toxic chemicals

The MPs are reported to leach toxic chemicals into the receiving environment with further implications. The additives such as plasticizers, stabilizers, etc. used in plastic manufacturing are the primary reason for the leaching of toxic chemicals from MPs in different environments (Chen et al., 2021). Likewise, the leaching of toxic chemicals from MPs in the AD process has proven to be an inhibitory factor (Fig. 3) (Mohammad Mirsoleimani Azizi et al., 2021; Wei et al., 2020). The organic additives such as phthalates, bisphenol A (BPA), heavy metals etc. have reported to affect the propagation and development of microorganisms (Wang et al., 2016; Qiu et al., 2021). The plastic additives have reported to cause toxic effect on *Amphibalanus Amphitrite* and *Nitocra spinipes* (Wei et al., 2019b). Among the several additives, BPA has found to be a highly used additive in PVC manufacturing with higher toxic abilities to the microorganisms (Suhrhoff and Scholz-Böttcher, 2016).

The PVC MPs notably release bisphenol-A (BPA) and affect the biochemical pathways of the AD process by reducing the microbial communities as well as their associated enzyme activities. The intensity of their toxicity is related to the abundance of PVC MPs in the AD matrix. Wei et al. (2019c) reported the BPA concentration of 1.8–3.6 μg/L inhibited all of the AD biochemical pathways wherein acidification inhibition by 6.9–16.8% and methanogenesis inhibition by 9.4–14.2% was observed along with the decreased protease and acetate kinase activities. In addition, the microbial communities such as *Rhodobacter*, *Proteiniborus*, *Garciella* and *Methanosaeta* were found to be reduced. Sharma and Melkania (2018) reported inhibition on acidification at the BPA concentration of 0.5–25 mg/L.

Hou et al. (2017) stated the hydrolysis inhibition by 7% at a BPA concentration of 20 mg/L caused by the inhibition of α-amylase activity at pH 7.4. pH found to be an influential factor in BPA leachability and its further impact on the hydrolysis process. Jiang et al. (2021) reported that the addition of 50 mg/kg TS of BPA enhanced protein hydrolysis at pH 10 while showing no impact on carbohydrate hydrolysis. Hence, the inhibition of BPA on hydrolysis could occur at neutral pH conditions. In contrast, the effect on hydrolysis from phthalates additives depends on the type and concentration rather than the AD operating conditions. DBP leached from PET MPs at the concentration of 10–60 particles/g TS has been shown to inhibit the hydrolysis of protein and carbohydrates at pH 10 (Wei et al., 2019b). The reduction of hydrolytic bacteria *Bacteroides*, *Tissierella* and *Fonticella* microbial communities were prominent. Recent studies have demonstrated that the DBP leached from 10 to 60 particles/g TS PET MPs inhibited the hydrogen production by 5.6–15.5% (Wei et al., 2019c). Similarly, the BPA at the concentration of 0.5–25 mg/L has shown the inhibition on hydrogen production by 9.2–75.3% at pH 5.5 (Sharma and Melkania, 2018).

Another commonly found MP PS is reported to leach out sodium dodecyl sulfate causing the inhibition on methane production (Wei et al., 2020). Wei et al. (2019b) reported that the leaching of BPA from PVC facilitated the destruction of microbial cells along with the solubilization of extracellular polymeric substances (EPS). The PVC exposure increased soluble chemical oxygen demand (sCOD) due to the cellular components released from the ruptured microbial cells and solubilized EPS. It is also reported that up to 80% removal of BPA in AD process could be achieved, hence, the concentration of BPA at different stages of AD process may have varied effect on the microbial community.

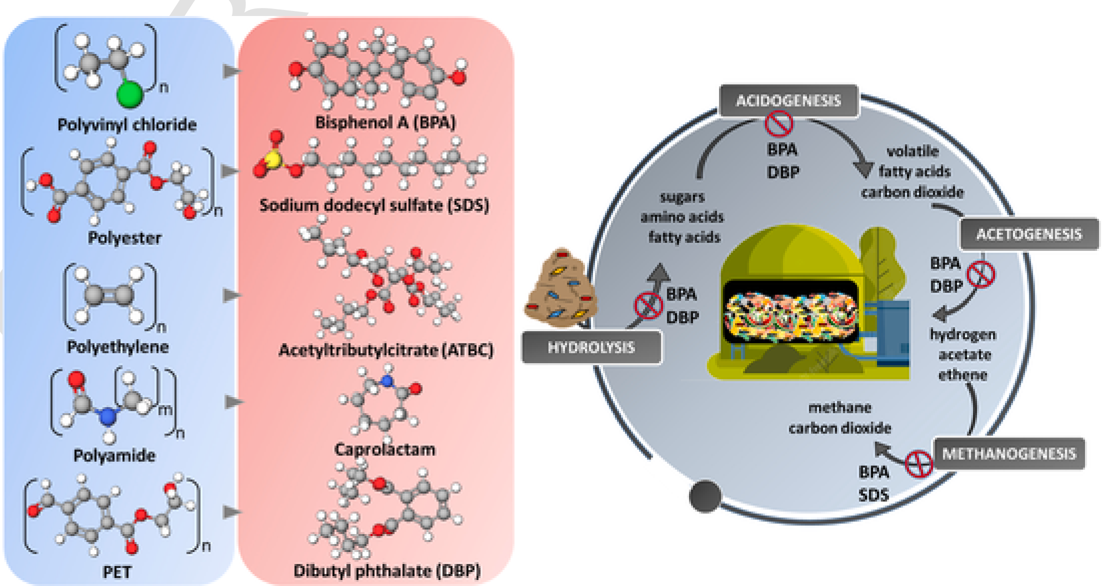

Fig. 3. Leaching of toxic compounds from the microplastics (MPs) affecting the biochemical pathways of AD process.

Hence, the operating conditions of AD process directly influence the impact of BPA on the microorganism involved in AD. Furthermore, few studies have reported no inhibition from the leached chemicals of MPs on the AD microbial community. Wei et al. (2019a) reported that the leaching additive acetyl tri-n-butyl citrate (ATBC) from PE showed no inhibitory effect on the methanogenic microorganisms. These studies provide early insight into the impact of toxic chemicals leached from MPs on different stages of AD process. Further investigation pertaining to the role of leached chemicals from different types of MPs at different concentrations on the anaerobic microbial communities is needed.

(b) ROS formation by MPs

The MPs are known to generate reactive oxygen species (ROS) such as H_2O_2, ·OH, ·O_2, etc. in different environmental conditions. The cellular enzymes and antioxidant compounds regulate the ROS concentrations and their production rates. Generally, microorganisms have the innate ability to remove the ROS by utilizing the antioxidant enzymes which reduces the oxidative stress. However, the excessive production of ROS may hinder the redox balance by activating redox-sensitive signaling pathways causing the toxicity and destruction of cells (Wei et al., 2021) During the biological treatment, ROS generation could be considered as an inhibitory factor since it directly affects the microbial viability (Fig. 4). For example, PP MPs could directly influence the cellular ROS concentrations (Hwang et al., 2019).

ROS generation from MPs under aerobic environment is well known, however, the recent studies have demonstrated the possibilities of ROS generation from MPs under the AD conditions as well (Mohammad Mirsoleimani Azizi et al., 2021). ROS generation induced from MPs exposure could be an important parameter to assess the MPs toxicity, cellular activities like growth and death. Hwang et al. (2019) conducted an experimental study to evaluate the PP induced toxicity on cells using ROS assay kit. PP particles (25 μm) increased the ROS generation by 30% wherein the larger particles did not show any difference indicating strong toxicity of smaller PP MPs on the cell viability. The direct contact of smaller PP particles resulted in the cellular uptake and increased the ROS generation by altering the biochemical equilibrium within the cells. Wei et al. (2020) reported that the PE and PS MPs significantly enhance the intracellular ROS levels causing the destruction of microbial cells. Wei et al. (2019a) investigated the ROS generation from PE MPs during sludge AD process. The intracellular ROS production at PE concentration of 100 and 200 particles/g TS was directly proportional to the cell viability as the viability decreased from 92.4% to 84.6%. The larger surface area of MPs assists in the catalytic reaction between the exposed reactive groups of MPs with the molecular dioxygen causing the ROS generation. Further, the generated ROS triggers the inflammation and apoptosis pathways leading the cell death.

Battacharya et al. (2010) reported increased ROS production under PS (0.2 μm) exposure at concentration of 0.08–0.58 mg/L and affected the growth of *Chlorella* sp. and *Scenedesmus* sp. Wei et al. (2021) studied the toxic effect of ROS produced from PET MPs using ROS scavenging assay with antioxidants. PET MPs induced the ROS generation by 49–52% compared to the control without PET MPs. Further simulation studies revealed that the oxidative stress caused by PET MPs were the primary reason for the degradation rate. During the presence of MPs under AD conditions, the ROS generation at the sub-micromolar concentrations of oxygen is possible, however, further research is required to understand the mechanism behind this pathway at different stages of AD process (Wei et al., 2019a).

5. Impact of microplastics (MPs) on antibiotic resistance genes (ARGs)

Apart from the inhibition of methane production by MPs, the changes in microbial communities due to the MPs stress affect the abundance of ARGs as they generally carried by different bacterial popula-

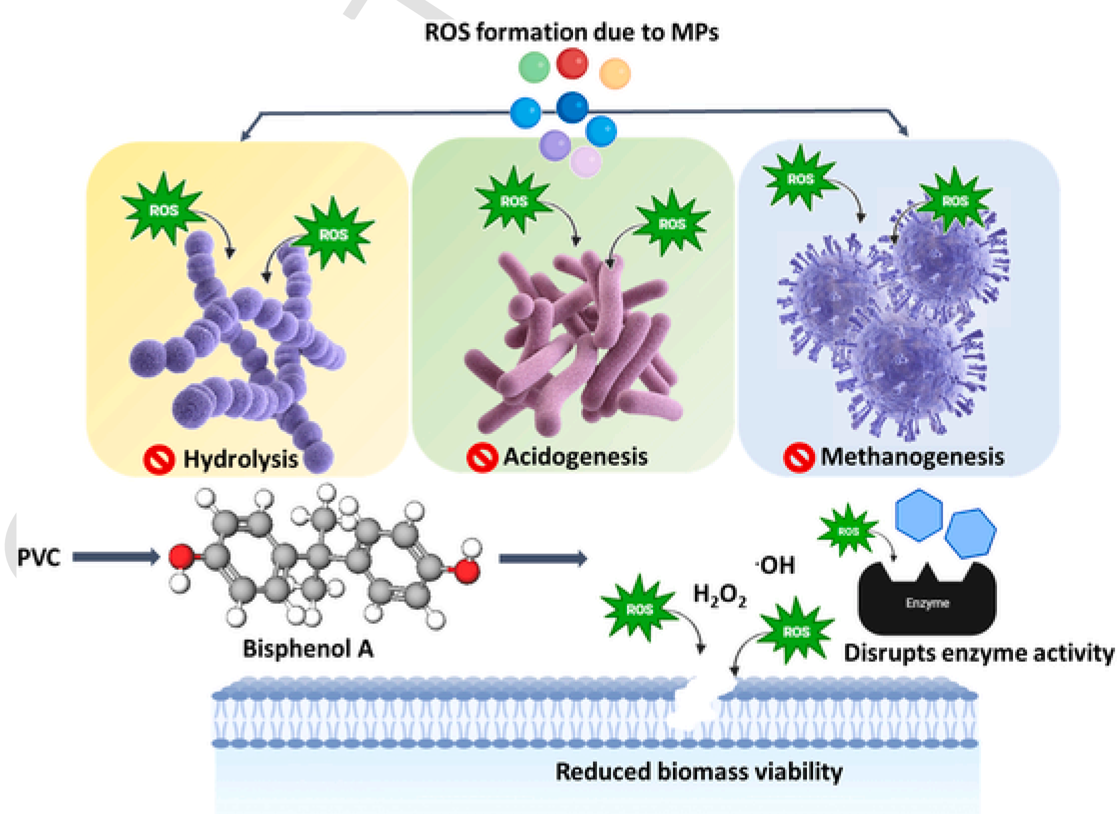

Fig. 4. Production of reactive oxygen species (ROS) from microplastics (MPs) and their subsequent effect on microbial cells during AD process.

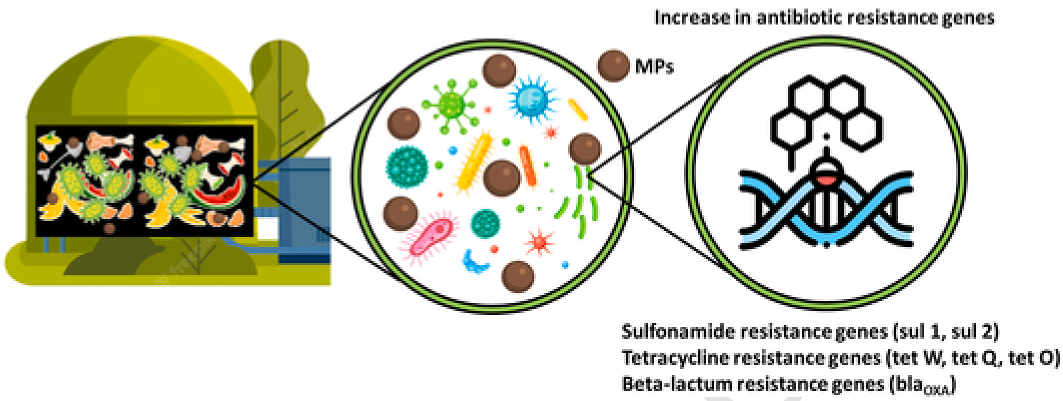

Fig. 5. Schematic representation depicting the enhanced abundance of ARGs due to the microplastics (MPs) stress during AD process.

tions (Fig. 5) (Haffiez et al., 2022; Shi et al., 2022). Recently, Shi et al. (2022) investigated the impact of PE MPs – 180 μm and PE MPs – 1 mm on different types of ARGs during sludge AD (Table 2). The results suggested that at the end of AD process, ARGs *sul2* had 7.6–9.6 × 10^8, bla_{OXA} was 1.3–2.2 × 10^7, *tetW* was 2–2.3 × 10^8, *tetO* was 6–8.8 × 10^8, and *tetM* was 1–1.4 × 10^7 copies/g sludge. Overall, the enrichment of ARGs by 1.2–3 times and 1.5–4 times in the presence of PE MPs – 180 μm and PE MPs – 1 mm, respectively was reported. This was attributed to the co-occurrence relationship between ARGs and bacterial species hosting ARGs in the presence of PE MPs. Wang et al. (2023) also reported an increase in the abundance of ARGs and mobile genetic material (MGEs) due to the presence of PE MPs ranging from 40 to 48 μm in AD process. The abundance of ARGs were enhanced by 0.41%–5.34%. It was reported that there was an increase in *Thermoanaerobacter* and *Caldicoprobacter* hosting ARGs by 5.2% and 28.03%, respectively. A strong positive correlation was observed between the PE MPs and the enrichment of ARGs and MGEs which suggests that MPs can aid in the propagation of both ARGs and MGEs during AD process. Similarly, Luo et al. (2022) reported an increase of 14.8% and 23.6% in the ARGs with PE and PVC MPs in the AD sludge process, respectively. The dominant species reported were *Acinetobacter* sp. and *Salmonella* sp. It was also reported by the authors that with PVC MPs in the digester stimulated the acquirement of ARGs by functional microbes and human pathogen bacteria which was not observed with PE. Zhang et al. (2021a, 2021b) reported that the abundances of *tetC*, *tetG* and *tetW* increased from 0.71 to 12.63 to 1.15–28.62 by altering the ARGs host bacterial profiles. In another study, Zhang et al. (2022a, 2022b, 2022c) reported that PET enhanced the abundance of 8 different ARGs (*ereA*, *ermF*, *tetM*, *tetG*, *sul1*, *sul2*, *mexF*, *tetS*) by 1.7–2.5 times along with the increase in host bacterial species. Overall, these studies

Table 2
Summary of studies on impact of microplastics (MPs) on ARGs during AD process.

MPs	Concentration	Waste type	ARGs	Observation	Reference
PE	200 particles/g TS	WAS	*sul2, tetW, tetQ, tetO*	ARGs increased by 1.2–4 times	Shi et al., 2022
PE	50-200 particles/g TS	WWTP sludge	*tetM, tetO, tetB, sul2, ErmF, ErmG*	Increased ARGs by 0.87%–5.34%	Wang et al., 2023
PE and PVC	30 particles/g TS	WA30S	*tetA, tetG, mphE, ant 3, ermG, msrE, ereA2, sul1, mtrA, catB8*	Increased ARGs abundance by 14.8–23.6%	Luo et al., 2022
PET	170 particles/g TS	Sewage sludge	*tetM, tetG, tetS, ereA, ermF, sul1, sul2, mexF*	The abundance of ARGs increased by 1.7–2.15 times	Zhang et al., 2022a, 2022b, 2022c

show the early progress on the detection and quantification of increased ARGs abundance due to the exposure of MPs during AD process.

6. Future perspectives

The comprehensive literature on the impact of MPs on the performance of AD process revealed that the MPs exposure significantly affected the process performance by directly hindering the microbial activities. The path of MPs entry into AD process revealed the necessary of improved organic waste collection systems without the MPs contamination. The currently available literature on the effect of MPs on AD process demonstrated the inhibitory effect on the different biochemical pathways of AD process. The mechanistic understanding revealed that MPs exposure could directly or indirectly affect the microbial community structure thereby altering the process performance. The direct mechanism involved physical contact of MPs and microbes causing the cell pitting and destruction of cells, whereas the attachment of pollutants such as pesticides, heavy metals etc. could also induce the toxic effect on the microbes. Furthermore, the indirect mechanism included leaching of toxic chemicals from the MPs which eventually inhibit the microbial activities causing the deprived process performance. In addition, ROS generation upon the exposure of MPs creates the oxidative stress and affects the cellular regulation resulting in the destruction of cells. Besides, recent studies have demonstrated the enhanced ARGs due to the exposure of MPs during AD process. Overall, the MPs were found to have significant impact of the AD process. However, the research studies on the assessment of MPs exposure in AD process is still limited and demands for an extensive research to determine the robust alternatives to avoid the MPs impact. Future studies should consider the physiological parameters of MPs such as shape, color, size etc. on the performance of AD process. Further, the concentration of MPs and mixture of MPs should also be considered to assess their impact in realistic situations at different stages of AD process. In addition, the accumulation of MPs after the AD process, i.e., in the digestate and its further treatment should receive attention as the digestate disposal will be the pathway for MPs to reenter the terrestrial eco system.

7. Conclusions

The MPs exposure on the AD process showed the evidence of inhibitory effect on the process performance through different mechanisms. In the current waste management system, the contamination of MPs in organic waste collection is inevitable, hence, mechanistic understanding of impact of MPs on AD process is necessary to develop remedial solutions with minimum impact on the process. Although the previous studies proved the negative impact of MPs on AD process, further detailed investigation is needed considering the varied nature of MPs exposure and organic waste characteristics. Besides, the available studies have been conducted using single MPS whereas the real conditions include exposure of multiple types of MPs on AD process which needs further investigation. In addition, the digestate disposal needs to be assessed to reduce the risk posed by the possible transmission of ARGs or pollutants through MPs into the terrestrial environment.

Author statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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